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A Novel Approach for High Frequency Interior Noise Prediction

Ravi (S. T.) Raveendra, Alexis Castel, Pinak Salvekar, and Bryce Gardner ESI Group

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Abstract

Since Statistical Energy Analysis (SEA) is based on lumped parameters, acoustic responses predicted by SEA are spatially discontinuous. However, in many practical applications, the ability to predict spatially continuous energy flow is useful for guiding the design of systems with improved acoustical characteristics. A new approach, utilizing integral equations derived from energy flow concepts, is developed to predict the continuous variation of acoustic field such as sound pressure level in the interior of acoustic domains using structural response predicted by SEA. The computer code developed based on energy flow boundary integral equations is initially validated by analyzing sound propagation in a duct. Subsequently, interior noise levels within simplified airplane cabin and generic cargo van models are predicted using SEA alone and the present hybrid approach that combines SEA and Energy Boundary Element Analysis (EBEA) to highlight the validity and utility of the present approach.

Introduction

Numerical methods are widely used for structural vibration and acoustic analysis in many industries. Finite Element and Boundary Element formulations are the most conventional tools for structural-acoustic simulation. However, at high frequencies, when the dimension of the structure is large with respect to the wavelengths, these methods become computationally expensive or even impractical. In addition, structural-acoustic systems are highly sensitive to changes in environmental or boundary conditions at high frequencies. In order to account for these variations, the averaged response with respect to frequency, space or across an ensemble is utilized in the solution of high frequency vibroacoustic problems. Statistical Energy Analysis (SEA) is a well-established approach for predicting the average response of structural-acoustic systems at high frequencies [1]. In SEA, a vibro-acoustic system is divided into subsystems of similar modes. The lumped averaged energy within each subsystem of similar modes comprises the primary SEA variable and the power transferred between subsystems is expressed in terms of coupling loss factors. SEA models result in few equations and are easy to solve. However, since SEA is based on lumped parameters, no information is available on the spatial variation of the energy within a subsystem, thus it is not conducive for predicting continuous spatial variation of acoustic response.

An alternative approach based on energy flow concept has been developed that conceptually allows the prediction of continuous variation of energy variables. Energy Flow Analysis (EFA) is based on deriving the governing energy flow differential equation in terms of space and frequency averaged energy density variables using thermal conductivity analogy [2]. In the early adaptation of energy flow concept, the governing equation of structural acoustic systems is solved using discretization methods such as finite element methods [3], [4], [5], [6], [7], [8], [9], [10], [11]. While the coupled structural acoustic systems can be solved using energy finite element methods, the subdivision of the acoustic space could be avoided by combining Energy Finite Element Analysis (EFEA) with Energy Boundary Element Analysis (EBEA) [12]. However, although many researchers have subsequently extended EFA methods for the solution of many practical problems [13], [14], [15], the widespread industrial use of the method is still lacking.

On the other hand, SEA is the primary approach used for the solution of high frequency vibroacoustic, practical problems in most industries. But, as previously indicated, SEA is not conducive for predicting continuous variation of sound field. Thus, noise prediction by combining SEA with EBEA [16] to predict spatially continuous acoustic fields within enclosures is expected to be very beneficial in many practical applications. A hybrid approach that utilizes the best features of SEA and EFA is presented here. The analysis is performed sequentially in that the vibroacoustic problem of interest is initially solved using SEA and the results are used as boundary conditions to calculate acoustic field such as sound pressure level at any desired location in the interior utilizing EBEA. The interior noise levels computed by the present hybrid method are compared to solutions obtained by using VA One [17], a commercially available software that has been extensively used and validated in many industries, to verify the proposed approach. In those applications, acoustic fields

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are also mapped on arbitrarily selected interior virtual microphone locations (data recovery meshes) to display the spatially continuous interior acoustic fields. The approach presented here is not available in the literature to the best of our knowledge.

Methodology

Similar to SEA, EFA is based on the concept that at high frequencies, the energy density and intensity are solutions of far-field pressure wave solutions and acoustic field is reverberant and incoherent. At steady state, the conservation of power over a differential acoustic volume can be expressed as:

$$\nabla\langle\vec{q}\rangle+\langle\Pi_{dis}\rangle=\langle\Pi_{in}\rangle \tag{1}$$

where $\langle\vec{q}\rangle$ is the time and space averaged energy intensity vector, and $<\Pi_{dis}>$ and $<\Pi_{in}>$ are time and space averaged dissipative and input power, respectively. For hysteretic damping, the dissipated power is related to energy density as [18]:

$$\langle\Pi_{dis}\rangle=\eta\omega\langle e\rangle \tag{2}$$

where $<e>$ is the time and space averaged energy density, η is the damping loss factor and ω is the frequency of harmonic excitation. Using thermal conductivity analogy, a relationship between time and space averaged energy intensity and energy density can be derived as [2]:

$$\langle\vec{q}\rangle=-\frac{c^2}{\eta w}\nabla\langle e\rangle \tag{3}$$

where c is the speed of sound of the acoustic medium.

The governing differential equation of energy flow in an acoustic medium is derived by substituting equations (2) and (3) into equation (1) as:

$$-\frac{c^2}{\eta\omega}\nabla^2\langle e\rangle+\eta\omega\langle e\rangle=\langle\Pi_{in}\rangle \tag{4}$$

or

$$\nabla^2\langle e\rangle-\left(\frac{\eta\omega}{c}\right)^2\langle e\rangle=-\frac{\eta\omega}{c^2}\langle\Pi_{in}\rangle \tag{5}$$

The above equation can be recast to a form similar to standard Helmholtz equation for low frequency acoustic analysis as [16]:

$$\nabla^2\langle e\rangle+\bar{k}^2\langle e\rangle=\langle\bar{\Pi}_{in}\rangle \tag{6}$$

where $\bar{k}=j\frac{\eta\omega}{c}$ is an equivalent complex wave number, $\bar{\Pi}_{in}=-\frac{\eta\omega}{c^2}\Pi_{in}$ is the modified input power, and $j=\sqrt{-1}$ is the complex constant. Equation (6) is similar to the Helmholtz equation for low frequency acoustic analysis; the difference being, in the present case the wave number is a complex value, whereas the low frequency wave number is real for real values of density and speed of sound of the acoustic medium. The integral equation corresponding to this modified Helmholtz equation is conventionally derived by using the Green function and divergence theorem [19]. The integral equation for acoustic energy density within the acoustic domain then becomes:

$$\langle e\rangle=\int_S\left\{\bar{G}\frac{\partial\langle e\rangle}{\partial n}+\langle e\rangle\frac{\partial\bar{G}}{\partial n}\right\}dS+\int_V\bar{\Pi}_{in}dV \tag{7}$$

where $\bar{G}$ is the modified fundamental solution, S is the boundary of the acoustic domain, V is the volume of the acoustic domain, and n is the outward normal at the surface. The modified fundamental solution is derived from the solution of Helmholtz equation due to unit source as [16]:

$$G=\frac{1}{4\pi r}\exp\left(-j\bar{k}r\right) \tag{8}$$

where r is the distance between the source and field points. Note that unlike the fundamental solution corresponding to conventional Helmholtz equation, which is expressed in terms of complex arithmetic, the fundamental solution of the modified Helmholtz equation is in terms of real arithmetic.

In typical situations, the power is applied in a controlled volume and zero elsewhere and therefore, there is no need to perform a domain discretization to compute the contribution of input power in equation (7). A numerical system of equation can be developed from the surface integral counterpart of equation (7) as [19]:

$$[A]\{e\}=\{b\} \tag{9}$$

where [A] is the system matrix which is derived from the second integral of equation (7), {e} is the vector of the surface acoustic energy density, and the right-hand side vector {b} is obtained from the normal energy intensity at the boundary derived from SEA. In the present development, a collocation approach is used to generate the matrix equation (9). Upon the evaluation of energy density at the surface by solving the matrix equation (9), interior values of energy density are computed using integral equation (7) which is then used to calculate other acoustic quantities of interest such as sound pressure level.

In the present analysis, the structural response is computed using SEA [17] and the energy density in the acoustic domain is obtained by solving equations (8) and (7). The consequence of the use of this hybrid approach is that the continuous variation of energy density and subsequently the energy intensity and sound pressure levels within acoustic cavity domains can be evaluated at any specified location.

Examples

Example problems are solved to demonstrate the applicability and utility of the present development. The first example problem was used to validate the implementation of energy flow boundary element analysis. The second and third example problems are used to show the advantage of the hybrid method for predicting continuous spatial variation of relevant acoustic quantities such as sound pressure level in acoustic cavities.

Rectangular Duct

A rectangular duct of cross section 0.2 m × 0.2 m and length of 1 m was used to validate the implementation of the energy

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FIGURE 1 Boundary element model of rectangular duct

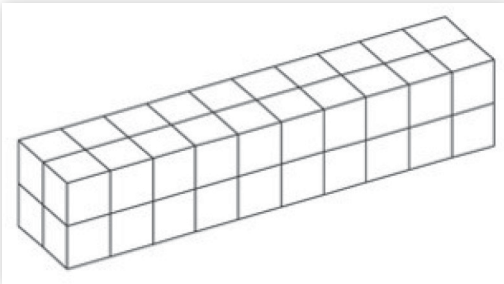

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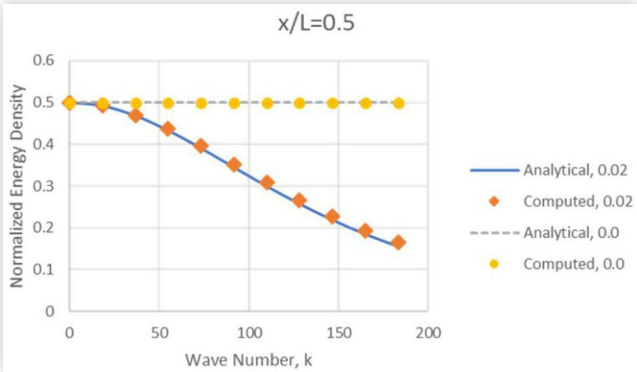
FIGURE 2 Comparison of SPL at the middle of the duct



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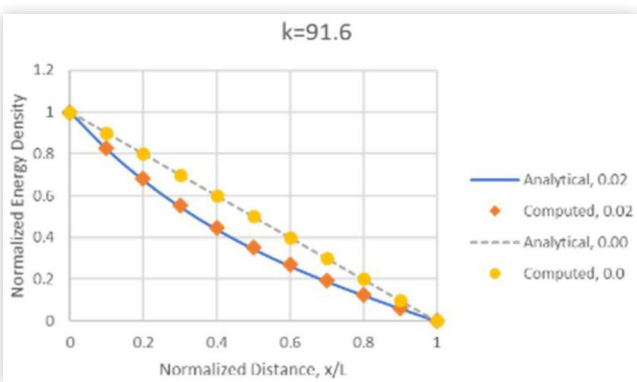
boundary integral method. A unit energy density was applied at the left end of the duct. Unlike acoustic pressure and velocity, energy density and intensity do not vary much spatially and therefore there is no need to use a fine mesh. The boundary element surface mesh used for the present calculation is shown in Figure 1.

In Figure 2, the analytical and computed frequency response functions of energy densities, calculated at the center of the duct and normalized with respect to applied energy density, are compared. Further, the spatial variation of normalized analytical and computed energy densities within the duct are compared for a frequency of 5000 Hz (wave number of 91.6) in Figure 3. The responses were calculated for damping loss factors of 0.0 and 0.02. The excellent agreement between the analytical and computed results indicate the validity of the implementation of energy flow boundary integral equations.

FIGURE 3 Comparison of spatial variation of SPL at k = 92



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Simplified Airplane Cabin

A simplified airplane cabin of length 3 m and diameter of 1 m was modeled using SEA. The SEA model comprised of 8 structural subsystems and 3 acoustic cavities as shown in Figure 4. A unit input power excitation (1 Watt) was applied at the center point of the bottom panel as shown in the figure. A damping loss factor of 0.01 was assumed for both structural and acoustic domains. The analyses were performed at the center frequencies of 1/3 octave band from 400 Hz to 10000 Hz. The results from SEA are used as boundary conditions to solve the energy flow integral equations. The boundary element mesh used is shown in Figure 5.

The sound pressure levels (SPL) predicted by the SEA and hybrid (SEA and EBEA) procedures at the center of the cabin are compared in Figure 6, indicating good correlation.

FIGURE 4 SEA structural and acoustic models of airplane cabin

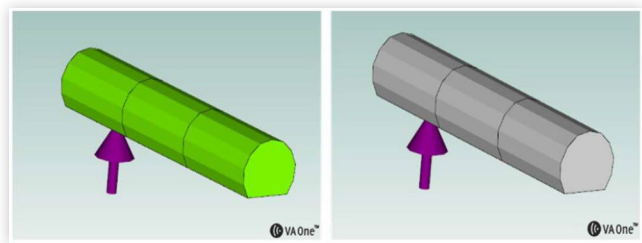

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FIGURE 5 EBEA model of airplane cabin

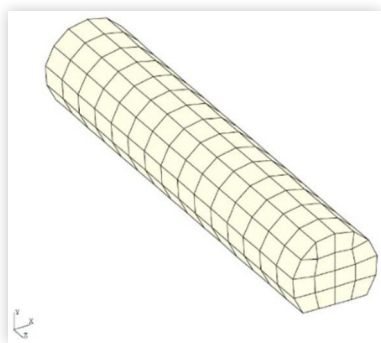

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FIGURE 6 Comparison of SPL at the center cavity

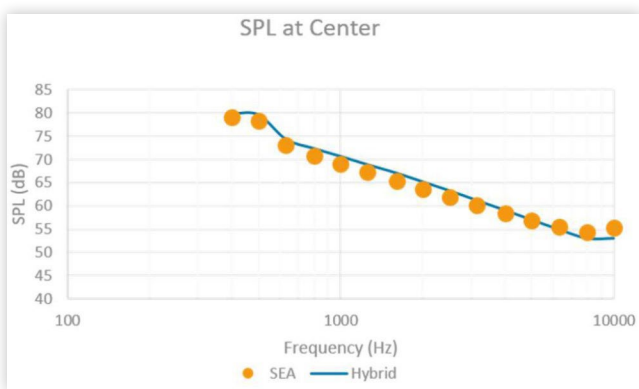

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FIGURE 7 SPL at the center of the cabin

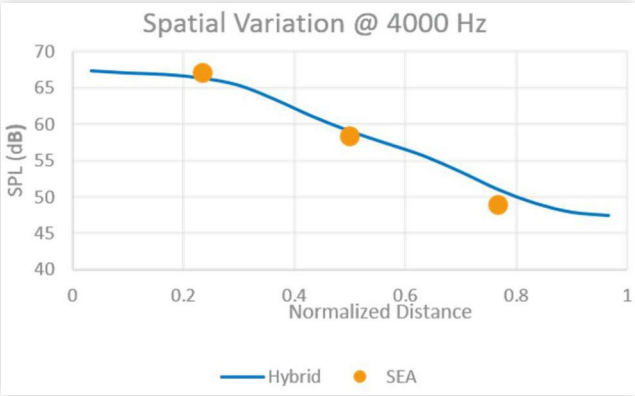

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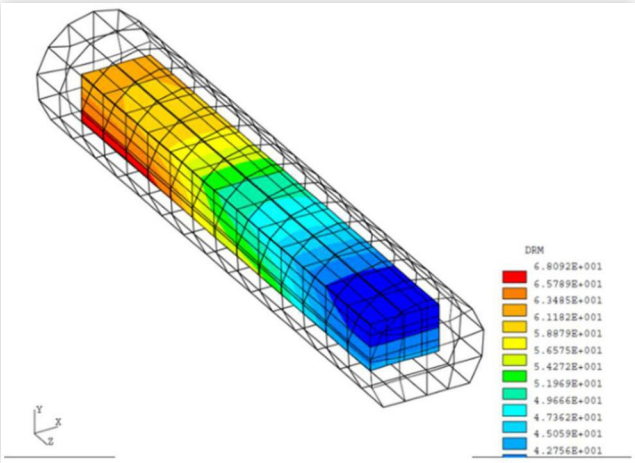
FIGURE 8 SPL on the data recovery mesh



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However, the utility of the hybrid method is the ability to compute the SPL at any interior location within the acoustic cavity. In Figure 7, the SPL predicted at 4000 Hz by the hybrid method is compared to the SEA results obtained at the center of the three cavities. These results again indicate good agreement between the approaches. A 'shaded' plot of the spatial SPL variation within the cavity on the data recovery mesh (field locations), predicted by the hybrid method, is shown in Figure 8 to highlight the efficacy of the hybrid method to predict spatially continuous acoustic field.

Generic Cargo Van

A cargo van was analyzed using both SEA and hybrid approaches. The acoustic domain of the cargo van was divided into engine cavity, driver cavity and passenger cavity by the dash and the bulkhead as shown in Figure 9. The SEA structural and acoustic models of the van are shown in Figure 10. A unit input power (1 Watt) was applied to the engine cavity and noise levels in the passenger cavity at the center frequencies of 1/3 octave band from 400 Hz to 10000 Hz were computed. The interior noise levels at the center of the front and back passenger cavities using SEA and hybrid methods are compared in Figure 11 and again good correlations are observed.

FIGURE 9 Cut-out-view of a generic cargo van

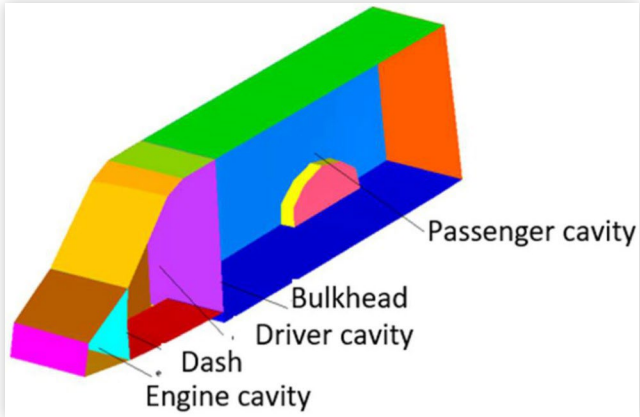

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FIGURE 10 SEA structural and cavity models of the van

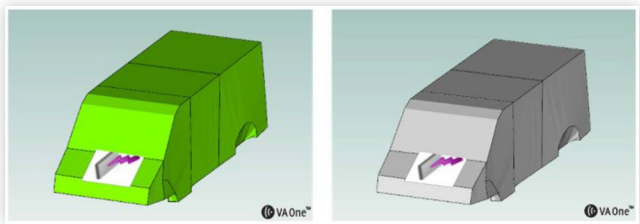

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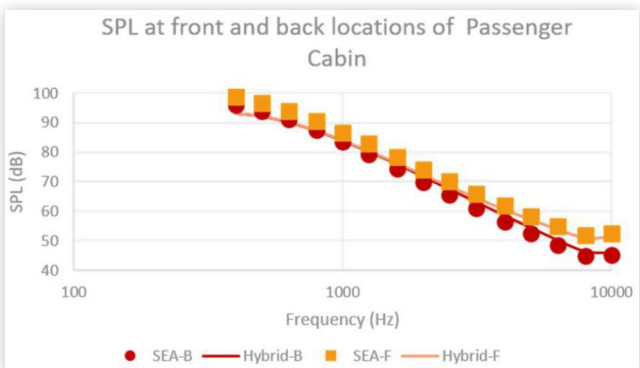
FIGURE 11 Variation of SPL at the front (F) and back (B) passenger compartment locations



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A comparison of the sound pressure variation at the front and back locations of the passenger cabin, predicted by the hybrid and SEA approaches is shown in Figure 12 for 1/3rd octave band center frequencies of 1000 Hz and 8000 Hz. These results are also in good agreement. Note that the boundary element mesh used in this example is the same size as the structural panel (subsystem) used in SEA as shown in Figure 10.

The utility of the hybrid method is the ability to predict the continuous spatial variation of the sound field at any location within the acoustic domain. The spatial variation of sound pressure levels at a frequency of 8000 Hz on the data recovery mesh, predicted by the hybrid approach, is shown in Figure 13.

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FIGURE 12 Spatial variation of SPL in the passenger compartment

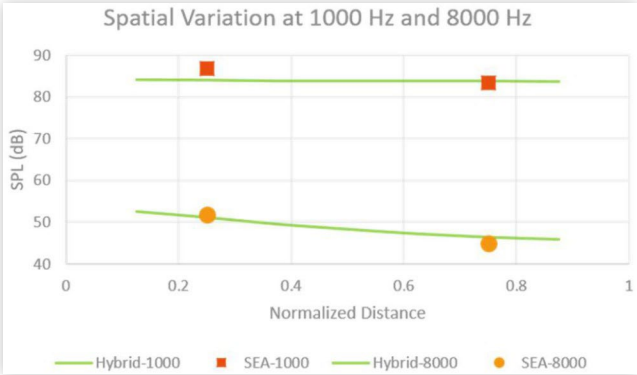

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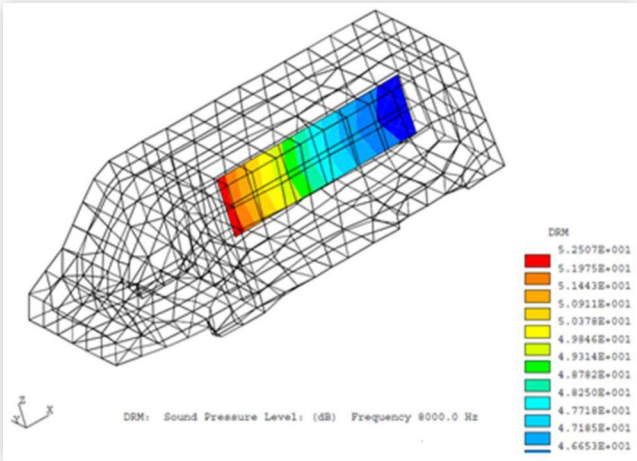
FIGURE 13 SPL variation in the passenger cabin - engine noise source



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The same van model, subjected to a unit power of 1 Watt applied to an external cavity next to the windshield as shown in Figure 14, was solved using SEA and hybrid approaches.

The frequency response at the center location of the passenger cabin calculated using SEA and hybrid approaches are compared in Figure 15, indicating excellent agreement between the predictions. The spatial variation of the sound pressure level in the passenger cabin for this loading condition,

FIGURE 14 SEA model with the exterior cavity

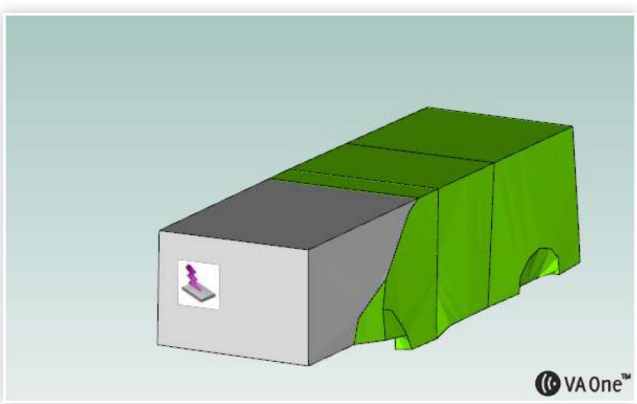

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FIGURE 15 SPL comparison at the center location of the passenger cabin

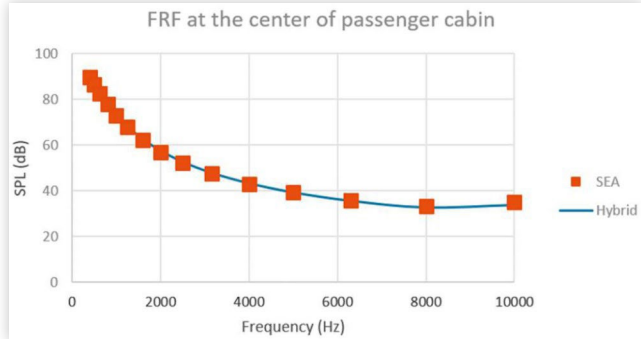

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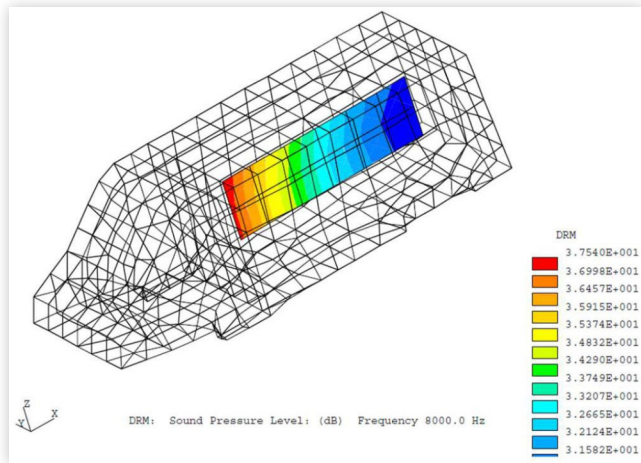
FIGURE 16 SPL variation in the passenger cabin - wind noise source



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computed using the hybrid approach, is shown in Figure 16 to demonstrate the advantage of hybrid method in predicting continuous variation of interior sound field.

Concluding Remarks

A hybrid method that combines statistical energy analysis and energy boundary element analysis is developed to predict the continuous spatial variation of noise levels at any location within acoustic cavities. The method is validated by solving example problems using SEA only and hybrid (SEA and EBEA) approaches. The observed results indicate good correlations between SEA and hybrid predictions. More importantly, the hybrid method allows the noise prediction at any specified location within the acoustic domain whereas SEA models assume constant energy within the acoustic subsystems and therefore does not predict the continuous variation of acoustic quantities within acoustic domains. It should be noted that the accuracy of the hybrid approach depends on the accuracy of the SEA since the boundary conditions of the acoustic analysis, utilizing EBEA, are derived from the SEA.

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However, the hybrid provides the means to predict continuous variation of the acoustic field that can be used to visualize energy flow and guide subsequent application of noise control procedures to improve the acoustical characteristics of vibroacoustic systems.

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Contact Information

Ravi (S. T.) Raveendra
ESI US R&D
32605 West 12 Mile Road, Suite 350
Farmington Hills, MI 48338
rrv@esi-group.com

Definitions/Abbreviations

SEA - Statistical Energy Analysis

EFA - Energy Flow Analysis

EFEA - Energy Finite Element Analysis

EBEA - Energy Boundary Element Analysis

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